

PYTHON FULL STACK + REACT JS + AI



Introduction To Python

- Why Python
 - Application areas of python
 - Features and Limitation of Python
 - Python versions
 - Installing python
 - Python interpreter architecture
1. Python byte code compiler
 2. Python virtual machine(pvm)

Writing and Executing First Python Program

- Using interactive mode
 - Using script mode
1. General text editor and command window
 2. Idle editor and idle shell
- Understanding print() function
 - How to compile python program explicitly

Python Language Fundamentals

- Character set
- Keywords
- Identifiers
- Comments
- Variables
- Literals
- Operators
- Reading input from console
- Parsing string to int, float

Python Conditional Statements

- If statement
- If else statement
- If elif statement
- If elif else statement
- Nested if statement

Looping Statements

- While loop
- While Else
- For loop
- For Else
- Nested loops
- Pass, break and continue keywords

Standard Data Types

- Int, float, complex
- Bool, nonetype
- Str, list, tuple
- Bytes, Bytearray, Range
- Dict, set, frozenset

String Handling

- What is string
- String representations
- Unicode string
- String functions, methods
- String repetitions and concatenation
- String indexing and slicing
- String formatting

Python List

- Creating and accessing lists
- Indexing and slicing lists
- List methods
- Nested lists
- Shallow v/s Deep Copy
- List comprehension

Python Tuple

- Creating tuple
- Accessing tuple
- Immutability of tuple
- Tuple to list
- List to tuple

Python Set

- How to create a set
- Iteration over sets
- Python set methods
- Union, intersection, difference of sets
- Python frozenset

Python Dictionary

- Creating a dictionary
- Dictionary methods
- Accessing values from dictionary
- Updating dictionary
- Iterating dictionary
- Nested Dictionaries
- Dictionary comprehension

Python Functions

- Defining a function
- Calling a function
- Types of functions
- Function v/s Method
- Function arguments
- Function return statement
- Nested function
- Function as argument

- Function return statement
- Nested function
- Function as argument
- Function as return statement
- Decorator function
- Closure
- Map(), filter(), reduce(), any() functions
- Anonymous or lambda function

Modules & Packages

- Why modules
- Script v/s module
- Importing module
- Understanding Pythonpath Variable
- Commonly Used Library Modules
- Why Packages
- Understanding __init__.py file
- Understanding pip utility

File I/O

- Why File handling
- File Pointer/Cursor
- File modes
- Functions and methods related to file handling
- Understanding with block

Object Oriented Programming

- Procedural v/s Object Oriented Programming
- OOP Principles
- Defining a Class & Object Creation
- Object class and it's methods
- Inheritance

Exception Handling

- Difference Between Syntax Errors and Exceptions
- Keywords used in Exception Handling
- 1. try , except , finally , raise , assert
- Types of Except Blocks

GUI Programming

- Introduction to Tkinter Programming
- Tkinter Widgets
- 1. Tk, Label, Entry, TextBox, Buttons
- 2. Frame, messagebox, filedialogetc
- Layout Managers
- Event handling
- Displaying image

Multi-Threading Programming

- Multi-processing v/s Multi-threading
- Need of threads
- Creating child threads
- Functions /methods related to threads

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Regular Expressions (Regex)

- Need of regular Expressions
- re Module
- Functions/Methods related to regex
- Meta Characters & Special Sequences

SQL

Introduction to RDBMS

- What is Relational Database Package
- Difference between SQL & Database
- Installing MySQL Server database

SQL Basic

- DDL: Create, Alter, Drop, etc.
- DML: Insert, Update, Delete ,etc.
- DQL : Select
- Autoincrement field
- SQL Comments
- SQL Aliases
- Savepoint & rollback

SQL Constraints

- Not NULL, Unique key
- Primary key, Check
- Default, Foreign key

SQL Operators

- Arithmetic operators
- Logical operators
- Conditional operators
- Like, between, in operators

SQL Clauses

- Order by
- Where
- Limit/top

- Group by
- having

SQL Clauses

- Order by
- Where
- Limit/top
- Group by
- having

SQL Joins

- Inner Join
- Left Join
- Right Join
- Full Join

SQL View

- creating view
- updating view
- fetching data from view

SQL Functions

- String functions
- Aggregate functions
- Date & time functions

Stored Procedures & Functions

- Understanding stored procedures and their key benefits
- Working with stored procedures
- Studying user-defined functions

Working with CSV Files:

- How to write result to csv file
- How to read csv file

Python Database Connectivity

- Database Drivers and connectors
- Creating connection object
- Understanding cursor object
- Executing SQL statements using cursor
- Fetching records from cursor
- Storing and retrieving Date and Time

Web Design

Introduction to WEB

- What are web applications?
 - Static v/s dynamic web applications
 - Web application Architecture
1. Front End
 2. Back End

HTML

- INTRODUCTION TO HTML
- headings
- paragraph
- formatting tags
- image
- anchor

- lists
- table
- ID
- class
- iframe
- div
- span
- form
- audio
- video
- html vs html5

CSS

Introduction to CSS

- ways to use CSS (inline, internal, external)
- css selectors
- css color
- border
- padding
- margin
- height
- width
- lists
- text
- display (none, inline, block, inline-block)
- flex
- grid
- position
- z-index
- media

Java Script

Js Basics

- keywords
- identifiers
- datatypes
- variables and constants
- tokens
- operators

Decision Making Statements

Conditions Statements Or Selection Statements

- if
- if else
- nested if else
- if else if ladder
- switch case

Iteration Statements Or Loops

- for
- while

- do while
- for in
- for of

Jump Statements

- continue
- break

Functions

- function declarations, denitions and calling
- types of functions
- anyonomous functions
- at arrow functions or arrow functions
- self invoked function (IIFE)
- storage classication or varibale scope (var let const and without keywords)
- recursion

Strings Methods

array Methods

Global Execution Context and Function execution context

- scope chain rule and lexical scoping
- closure

Objects

- literal and keys
- object functions
- nested objects
- Object Prototypes
- setter and getter methods

Set and Map

WeakMap and WeakSet

Dates and Time

OOPS

- literal and keys
- object functions
- nested objects
- Object Prototypes
- setter and getter methods

DOM Manuplation

- DOM
- BOM
- event handling
- time based events

Advanced Js

How Javascript Works

hoisting

strict mode

closure

synchronous and asynchronous programming

Event propagation

bubbling

capturing

Event loop

function curring

ajax and json

calling api using ajax

callback hell

promise

async await

fetch

modules

version

Typescript

- Why Typescript
- Basic Types
- Class and Interfaces
- Modules

Django

Introduction to Django

- Introduction to Back-End Web Development using Django
- HTTP protocol
- MVC model
- Virtual environment
- Django structure
- Generic Views
- HTML templates
- URL dispatcher

Advanced Django for Web and Automation

- Custom Views
- GET and POST methods
- URL shortener
- User model
- Logic in templates
- Querying models
- Serving Staticfiles
- Deployment of Django
- Automating tasks with Django

Django REST

Building Web APIs using Django REST

- Generic Views in Django Rest
- Serializers
- JSON
- Building RESTful APIs
- Filtering Models

- Working with Images
- Authentication with tokens
- Postman
- Related models
- Contenttypes app

Deploying Web APIs

- Using API endpoints
- Deployment of Django REST project

React Js

INTRODUCTION TO REACT JS

- What is React JS?
- What is SPA?
- DOM vs Virtual DOM
- Advantages and Disadvantages
- Key Features

ENVIRONMENTAL SETUP

- Node | NPM
- Installation of CLI
- Setup Project
- Directory Structure
- Code Editors
- How React JS Application Boot

BASIC FEATURES OF REACT JS

- React Concepts
- JSX and TSX
- Render Elements
- Function and Class Components
- Props and State
- Handling Events
- Dynamic Data Rendering
- Property Binding

KEY FEATURES OF REACT JS

- Conditional Rendering
- List and Keys
- Forms Handling
- Forms Validations

COMPONENT LIFECYCLE HOOK

- Understanding component lifecycle
- All Lifecycle Hooks

EVENT HANDLING REACT

- Understanding React Event System
- Passing arguments to event Handlers

NETWORK CALL

- Fetch
- Axios

CUSTOM SERVICES

- Introduction to Services
- Building a Service

LOCAL DATA STORAGE

- Local Storage
- Session Storage
- Cookies

ROUTING WITH REACT ROUTER

- Setting up React Router
- Configuring route with Route Component
- Making routes dynamic with Route Params
- Working with nested routes
- Link and NavLink
- Redirect Routes

UI COMPONENTS

- Material Design
- PrimeNG

INTRODUCTION TO REDUX

- Why Redux
- Install and setup
- Store, Reducer, actions
- Dispatcher
- High order Components
- mapStateToProps and map DispatchToProps usage

ADVANCE REDUX

- Async Actions
- Middleware
- Redux Thunk and Redux Saga

React Hooks

- Why We Need Hooks.
- Different Types Of Hooks
- Using State And Effect Hooks
- Usereducer, Useref Etc.
- Custom Hooks
- Rules Of Hooks

Third Party Modules

- Social Login
- Pagination
- Search
- Filter
- JWT Token
- File Upload
- Many More



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